1. INTRODUCTION

1.1 Project Overview

The project “Import Data using Transform Maps (Spreadsheet)” demonstrates how employee information can be imported into ServiceNow in a structured and reliable way. The spreadsheet contains fields such as Employee ID, Name, Email, Department, and Location. When the spreadsheet is uploaded, the data is first stored temporarily in an Import Set Table. A Transform Map then maps the source columns to the matching fields in the target Employee table.

Employee ID is used as the unique identifier to determine whether an employee record already exists. If the ID matches an existing record, the record is updated; if no matching record exists, a new employee record is inserted.

1.2 Purpose

• To reduce manual entry of employee information.

• To transfer spreadsheet data into ServiceNow in an organized way.

• To map spreadsheet columns to the correct target-table fields.

• To validate imported values and reduce data-entry errors.

• To update existing employees and insert new employees using Employee ID.

2. IDEATION PHASE

2.1 Problem Statement

Entering employee details manually into ServiceNow can take time and may lead to duplicate records, typing mistakes, or inconsistent field values. The project addresses this problem by using a spreadsheet, Import Sets, and Transform Maps to move employee information into the target table with less manual effort.

2.2 Empathy Map Canvas

The main users are HR staff, administrators, and employees responsible for maintaining employee records. Their needs include quick data entry, accurate information, easy correction of errors, and confidence that existing records will not be duplicated.

• **Says:** “I need to import many employee records at once.”

• **Thinks:** “The data should go into the correct fields without creating duplicates.”

• **Does:** Prepares a spreadsheet, uploads it, checks the import results, and corrects invalid values.

• **Feels:** Wants the process to be simple, dependable, and time-saving.

2.3 Brainstorming

Possible approaches considered include manual entry, direct spreadsheet upload, and using ServiceNow Import Sets with Transform Maps. The selected approach uses Import Sets for temporary storage and Transform Maps for field mapping and record processing. Employee ID acts as the matching key for update-or-insert behaviour.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User activity	Expected outcome
Prepare	Create spreadsheet with required employee columns	Consistent source data

Stage	User activity	Expected outcome
Upload	Upload spreadsheet into ServiceNow	Data reaches Import Set Table
Transform	Run Transform Map and field mapping	Columns map to target fields
Validate	Review transformation results and errors	Incorrect data can be corrected
Verify	Check target Employee table	Existing records updated; new records inserted

3.2 Solution Requirement

• ServiceNow instance with access to Import Sets and Transform Maps.

• A spreadsheet file containing Employee ID, Name, Email, Department, and Location.

• An Import Set Table to temporarily store the uploaded rows.

• A target Employee table with corresponding fields.

• A Transform Map configured to map source columns to target fields.

• Coalesce or equivalent matching configuration on Employee ID, where supported by the table setup.

• Permission to upload data, run the transform, and view the results.

3.3 Data Flow Diagram

Step	Process	Data / output
1	Spreadsheet upload	Employee ID, Name, Email, Department, Location
2	Import Set Table	Temporary staging records
3	Transform Map	Source-to-target field mapping
4	Validation and matching	Check required values and Employee ID
5	Employee Table	Insert new records or update matching records

Flow: Spreadsheet → Import Set Table → Transform Map → Field Mapping and Validation → Employee Table → Update Existing Employee / Insert New Employee.

3.4 Technology Stack

• **Platform:** ServiceNow.

• **Data source:** Spreadsheet (Excel/CSV, depending on the configured import method).

• **Import mechanism:** Import Sets.

• **Transformation:** Transform Maps and field mapping.

• **Target storage:** Employee table (the configured target table).

• **Matching logic:** Employee ID used as the unique matching identifier.

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The main problem is the effort and risk involved in entering multiple employee records manually. Import Sets and Transform Maps fit this need because they provide a repeatable process for loading rows, mapping fields, and processing records. Matching by Employee ID helps prevent the same employee from being inserted as a duplicate, provided the matching configuration and source data are correct.

4.2 Proposed Solution

The user prepares the spreadsheet using the required column names and uploads it to ServiceNow. The platform stores the uploaded rows in an Import Set Table. A Transform Map connects each source column to the appropriate target field. During transformation, Employee ID is used to match records. Matching records are updated, while unmatched records are inserted. The user then reviews the transform status and target table to confirm the outcome.

4.3 Solution Architecture

Source	Processing	Destination
Spreadsheet	Import Set Table	Temporary imported rows
Temporary rows	Transform Map	Mapped employee fields
Mapped data	Validation + Employee ID matching	Update existing / insert new
Processed records	Employee Table	Final employee records

Architecture summary: Spreadsheet → Import Set Table → Transform Map → Field Mapping & Validation → Employee Table. Employee ID determines whether a record is updated or inserted.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities
1. Requirement collection	Identify employee fields and target-table requirements.
2. Spreadsheet preparation	Create sample data with Employee ID, Name, Email, Department, Location.
3. Import configuration	Upload spreadsheet and confirm Import Set Table fields.
4. Transform configuration	Create field mappings and configure Employee ID matching.
5. Testing	Test new records, existing records, missing values, and invalid data.
6. Verification	Review transform logs and confirm records in the target table.

Suggested schedule: requirement collection (Day 1), spreadsheet preparation (Day 2), import and transform setup (Days 3–4), testing (Day 5), and verification/documentation (Day 6).

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing should confirm that the import process produces the expected results. Actual performance depends on spreadsheet size, instance configuration, network conditions, and transform rules. Record the measured import duration and processed-row counts during the demonstration.

Test case	Input condition	Expected result
New employee	Employee ID not present	A new employee record is inserted
Existing employee	Employee ID already present	The matching record is updated
Duplicate source ID	Same Employee ID appears twice	Behaviour is checked and handled according to configured rules
Missing required field	Name or required value is blank	Row is reviewed or handled by validation rules
Invalid email	Email value has invalid format	Validation identifies the value if validation is configured
Larger spreadsheet	Multiple rows uploaded	Rows process within the observed time and system limits

Performance measures to record: number of rows uploaded, number of rows transformed successfully, number of rows with errors, total processing time, and any duplicate or rejected records.

7. RESULTS

7.1 Output Screenshots

Add screenshots from your own ServiceNow instance in this section. Use the following suggested captions:

• Figure 7.1: Employee spreadsheet containing Employee ID, Name, Email, Department, and Location.

• Figure 7.2: Uploaded data displayed in the Import Set Table.

• Figure 7.3: Transform Map with source-to-target field mappings.

• Figure 7.4: Transform history or import log showing processing results.

• Figure 7.5: Employee table showing a newly inserted record.

• Figure 7.6: Employee table showing an existing record updated using Employee ID.

Result statement: The expected result is that spreadsheet rows are transferred to the target Employee table through an Import Set and Transform Map. Existing employees are updated and new employees are inserted according to Employee ID matching. Replace this expected-result statement with observed results after running the project.

8. ADVANTAGES & DISADVANTAGES

Advantages

• Reduces repetitive manual data entry.

• Supports importing multiple employee records in one process.

• Provides a clear staging and transformation workflow.

• Field mapping helps place values in the correct target fields.

• Employee ID matching can reduce duplicate employee records.

Disadvantages

• Incorrect spreadsheet headers or values may cause import errors.

• Transform Maps must be configured correctly.

• Poor source data quality can affect the final records.

• Large imports may take longer depending on instance resources and rules.

• Access permissions and table configuration are required.

9. CONCLUSION

The project demonstrates a structured way to import employee information into ServiceNow using spreadsheets, Import Sets, and Transform Maps. The Import Set Table stores incoming data temporarily, while the Transform Map maps fields and processes records into the target Employee table. Using Employee ID as the matching identifier supports updating existing employees and inserting new employees. The process reduces manual work and can improve consistency when the spreadsheet and transformation rules are configured correctly.

10. FUTURE SCOPE

• Add stronger validation for email format, mandatory fields, and allowed department values.

• Generate a summary report of inserted, updated, skipped, and failed rows.

• Schedule recurring imports for regular employee-data updates.

• Add error notifications and a correction workflow for rejected rows.

• Support additional source formats or integrate with an approved HR data source.

• Apply role-based access and audit logging for sensitive employee information.

11. APPENDIX

Source Code (if any)

This project primarily uses ServiceNow configuration (Import Set Table, Transform Map, field mappings, and coalesce/matching settings). Add any scripts or transform scripts here if they are used.

Dataset Link

Sample dataset: create or attach a spreadsheet with the columns Employee ID, Name, Email, Department, and Location. Insert your dataset URL here: _____.

GitHub & Project Demo Link

GitHub repository: https://github.com/srimugij-19/Import-Data-using-Transform-Maps-Spreadsheet-

Project demo link: _____

Note: Replace the blank student details and links, and insert genuine output screenshots from your ServiceNow instance before submission.